Course Outline

Course Name: Bioinformatics and Cloud Computing for Advanced Data Integration and Management (BINF5506)

Academic Year: 2025-2026

Faculty:

Program Coordinator: Fathiya Mohamed

Associate Dean: Erin Mandel-Shorser, PhD

Schedule Type: LEC

Land Acknowledgement

Humber College is located within the traditional and treaty lands of the Mississaugas of the Credit. Known as Adoobiigok [A-doe-bee-goke], the "Place of the Black Alders" in Michi Saagiig [Mi-Chee Saw-Geeg] language, the region is uniquely situated along Humber River Watershed, which historically provided an integral connection for Anishinaabe [Ah-nish-nah-bay], Haudenosaunee [Hoeden-no-shownee], and Wendat [Wine-Dot] peoples between the Ontario Lakeshore and the Lake Simcoe/Georgian Bay regions. Now home to people of numerous nations, Adoobiigok continues to provide a vital source of interconnection for all.
For more information, visit the Aboriginal Resource Centre (LRC2137) North Campus, (WEL301) Lakeshore Campus or www.humber.ca/aboriginal/

Faculty	Faculty of Health Sciences and Wellness
Program	Graduate Certificate in Clinical Bioinformatics, CB511
Course Name:	Bioinformatics and Cloud Computing for Advanced Data Integration and Management
Pre-Requisite(s)	N/A
Co-Requisite(s)	N/A
Equates	None

Restrictions	This course is restricted to Clinical Bioinformatics Graduate Certificate Students
Credit Value	3
Total Course Hours	42

Developed by: Sohail Hanif

Approved by: Erin Mandel-Shorser, PhD

Course Description

This course provides an introduction to the use of cloud computing in bioinformatics. Learners will explore cloud computing architecture and services, focusing on data curation, management and integration across platforms. The course covers workflow management systems and containerization to support learners in developing reproducible programs in bioinformatics.

Course Rationale

This course provides students with essential skills in cloud platforms like AWS for scalable infrastructure, data storage, and management solutions specific to bioinformatics needs. Students will learn how containerization technologies like Docker enhance reproducibility and portability of computational environments. The course also emphasizes scalable computing and parallel processing techniques in the cloud to handle large-scale bioinformatics data efficiently. Additionally, the course covers data curation, harmonization, and integration across platforms to ensure data quality and interoperability. By the end of the course, students will have developed the skills necessary to design and implement cloud-based bioinformatics workflows, manage and analyze large datasets, and apply advanced data management techniques—essential competencies for a career in modern bioinformatics research and applications.

Program Learning Outcomes Emphasized in this Course:

1. Identify relevant evidence-based research, trends, technologies, and standards within the field of bioinformatics to maintain currency in the industry.
2. Apply bioinformatics and computational methods to the design, execution and interpretation of scientific research.
3. Employ programming language and scripting versatility to be able to automate processes at the application level.
4. Examine bioinformatics data to discover patterns, draw conclusions and generate predictions for subsequent experiments. Complete complex genomic and proteomics applications using advanced principles of cell and molecular biology, pharmacology and toxicology, pathophysiology, endocrinology, and biochemistry.

Course Format(s)

LECTURE, ONLINE, DISCUSSIONS

Course Learning Outcomes

OQF Category	**At the successful completion of the course, the student will have demonstrated an ability to:**
Depth and Breadth of Knowledge	1. Compare and contrast different cloud computing architectures and services, focusing on platforms like AWS and Google Cloud, to understand their applications in bioinformatics data management.
Knowledge of Methodologies	2. Evaluate containerization technologies like Docker and workflow management systems such as Snakemake to assess their impact on reproducibility and scalability in bioinformatics analyses. 3. Design cloud-based bioinformatics workflows using scalable computing and parallel processing techniques.
Application of Knowledge	4. Apply cloud computing tools and services to manage, store, and analyze large-scale bioinformatics datasets. 5. Integrate data curation, harmonization, and integration techniques across platforms to ensure data quality and interoperability
Communication Skills	6. Utilize version control systems like Git to collaborate effectively on bioinformatics projects and document changes.

	7. Communicate complex bioinformatics cloud architectures effectively through appropriate visualizations and reports.
Professional Capacity/ Autonomy	8. Evaluate the impact of cloud computing tools and methods on genomics research advancements and their applications.

Assessment Weighting

Assessment	**Weight (%)**
Labs (5)	40
Projects (2)	40
Tests (2)	20

Modules of Study

Module	**Course Learning Outcomes**	**Assessments**
Cloud Computing Architectures and Services	1, 3, 4, 5, 7	Labs Project 1 Test 1
Version Control Systems	3, 6	Labs Project 1 Test 1
Workflow Management Systems	2, 3, 5, 6	Labs Project 1 Test 1
Docker and Containerization	2, 3, 5, 6	Labs Project 2 Test 2
Data Curation and Integration Across Platforms	1, 3, 5, 6	Labs Project 2 Test 2

Required Resources, Tools and/or Equipment:

- NONE

Supplemental Resources:

- Blackboard lectures and assigned resources

Additional Tools and Equipment

- Blackboard lectures and assigned resources

Prior Learning Assessment and Recognition (PLAR)

Students who have prior learning in the material of this course may be eligible for a course credit in recognition of their prior learning. The following table indicates the method that is used to assess prior learning for this course, or it indicates that such an assessment is not available. Students must apply for consideration for a prior learning assessment through the Office of the Registrar, and there is usually a fee associated with the application.

Portfolio	Challenge Exam	Skills Test	Interview	Other (Specify)	Not Available For PLAR
☑	☐	☐	☐	☐	☐

Policies and Procedures

It is the student's responsibility to be aware of their obligations under [Humber Policies and Procedures.]

Academic Regulations

It is the student's responsibility to be aware of the College Academic Regulations. The Academic Regulations apply to all applicants to Humber and all current students enrolled in any program or course offered by Humber, in any location. Information about academic appeals is found in the **Academic Regulations**.

Anti-Discrimination Statement

At Humber College, all forms of discrimination and harassment are prohibited. Students and employees have the right to study, live and work in an environment that is free from discrimination and harassment. If you need assistance on concerns related to discrimination and harassment, please contact the **Centre for Human Rights, Equity and Inclusion** or the **Office of Student Conduct**.

Accessible Learning Services

Humber strives to create a welcoming environment for all students where equity, diversity and inclusion are paramount. Accessible Learning Services facilitates equal access for students with disabilities by coordinating academic accommodations and services. Staff in Accessible Learning Services are available by appointment to assess specific needs, provide referrals and arrange appropriate accommodations. If you require academic accommodations, contact:

Accessible Learning Services

North Campus: (416) 675-6622 X5090

Lakeshore Campus: (416) 675-6622 X3331

Academic Integrity

Academic integrity is essentially honesty in all academic endeavors. Academic integrity requires that students avoid all forms of academic misconduct or dishonesty, including plagiarism, cheating on tests or exams or any misrepresentation of academic accomplishment.

Disclaimer

While every effort is made by the professor/faculty to cover all material listed in the outline, the order, content, and/or evaluation may change in the event of special circumstances (e.g. time constraints due to inclement weather, sickness, college closure, technology/equipment problems or changes, etc.). In any such case, students will be given appropriate notification.

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